Poonam Mehan

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linkedin github portfolio

Work Experience

Full Stack AI Engineer *Newral Softwares* *April 2026 - Present*

- **Real-Time Examination Platform**
  - Built a full-stack high-concurrency exam system sustaining **50K–80K concurrent users**; **NestJS/TypeScript** backend with a **GraphQL** API and **PostgreSQL/TypeORM**.
  - Engineered a **Redis** answer write-buffer and **BullMQ** worker pipeline (grading, checkpointing, delayed auto-submit) to decouple DB write load from concurrent-user count — keeping DB volume a function of server count, not exam size.
  - Secured with **JWT** auth and **SSE** real-time delivery; built student and admin **React** frontends (Apollo Client, Ant Design) against a shared GraphQL API.
- **Document-to-Quiz Engine**
  - Built an automated document-to-quiz engine (NestJS/GraphQL) converting uploaded Word question banks into structured, auto-graded assessments.
  - Implemented presigned **direct-to-S3** uploads (stateless backend, no local file storage), **Pandoc** DOCX→HTML conversion, **MathJax** MathML/LaTeX→SVG equation rendering, and **Cheerio**-based table parsing.
  - Enforced **multi-tenant RBAC** and a security-by-design answer-key split keeping correct answers server-side.
- **AI Page-Builder Microservice**
  - Architected an AI page-builder microservice (Node/Express, Bull + Redis, Socket.IO) generating course landing pages from prompts via **Gemini/Claude** (Vercel AI SDK).
  - Ran AI-generated HTML in isolated **E2B sandboxes** with live preview and real-time **WebSocket** progress.
  - Integrated with a NestJS/GraphQL backend (S3 publishing, multi-tenant AI-credit metering) plus **React/Redux** admin and public-facing frontends, bridging generated-page CTAs into a **Cashfree** payment flow.

Full Stack Engineer (Internship) *100xDevs* *July 2025 - December 2025*

- Built and shipped a Centralized Exchange Trading platform

SWE Fellow *Headstarter AI* *July 2024 - September 2024*

- Built and shipped multiple AI-powered and MERN full-stack projects over an intensive software engineering fellowship.
- Completed the program under mentorship from engineers at various YC-backed companies.

Projects

OpenLoop *Screen-context AI memory* **Swift, Electron, MCP, LiveKit, Gemini API**

- Mac menu bar app that reads apps via the macOS accessibility tree and exposes memory to Claude over MCP
- Gemini extraction with PII redaction and Zod-validated JSON, stored in encrypted SQLite with FTS5 search.
- Building a real-time LiveKit voice agent that answers from memory and confirms before acting on loops.
- Designed an LLM eval suite scoring commitment and fact extraction, memory updates and search accuracy.

My Legal Assistant *RAG Application* **MERN, Pinecone, Mistral AI**

- Full Stack RAG Application for Indian legal Q&A.
- Uses Pinecone for vector retrieval and Mistral AI for response generation.
- JWT based authentication and MongoDB backed chat history.

FlowBolt *Visual workflow automation platform* **Next.js, Express, Kafka, Websockets**

- Visual workflow automation platform connecting apps, services, and databases.
- Automates repetitive tasks with zero lines of code, no coding required.
- Built with Next.js, Express, Kafka, and WebSockets for real-time execution.

Asimo *AI-powered app builder* **Docker, React, AWS, XTerm JS**

- Builds full web apps from a single prompt, like Bolt or Lovable.
- Generates code from an LLM and runs projects in-browser via WebContainer.
- Deployed on AWS with a CI/CD pipeline, Dockerfile, and integrated terminal.

Education

B.Tech. Computer Science and Engineering *Guru Nanak Dev University, Amritsar* *2022-2026*

Technical Skills

JavaScript, TypeScript, React, Tailwind, Next.js, Turborepo, Express.js, Node.js, PostgreSQL, Pinecone, Figma, Wireframing, Git, Docker, AWS (EC2), WebSockets, Prisma